



REAL DATA. REAL INSIGHTS. A SAFER TOMORROW.

SOBZ EARTHQUAKE ANALYTICS

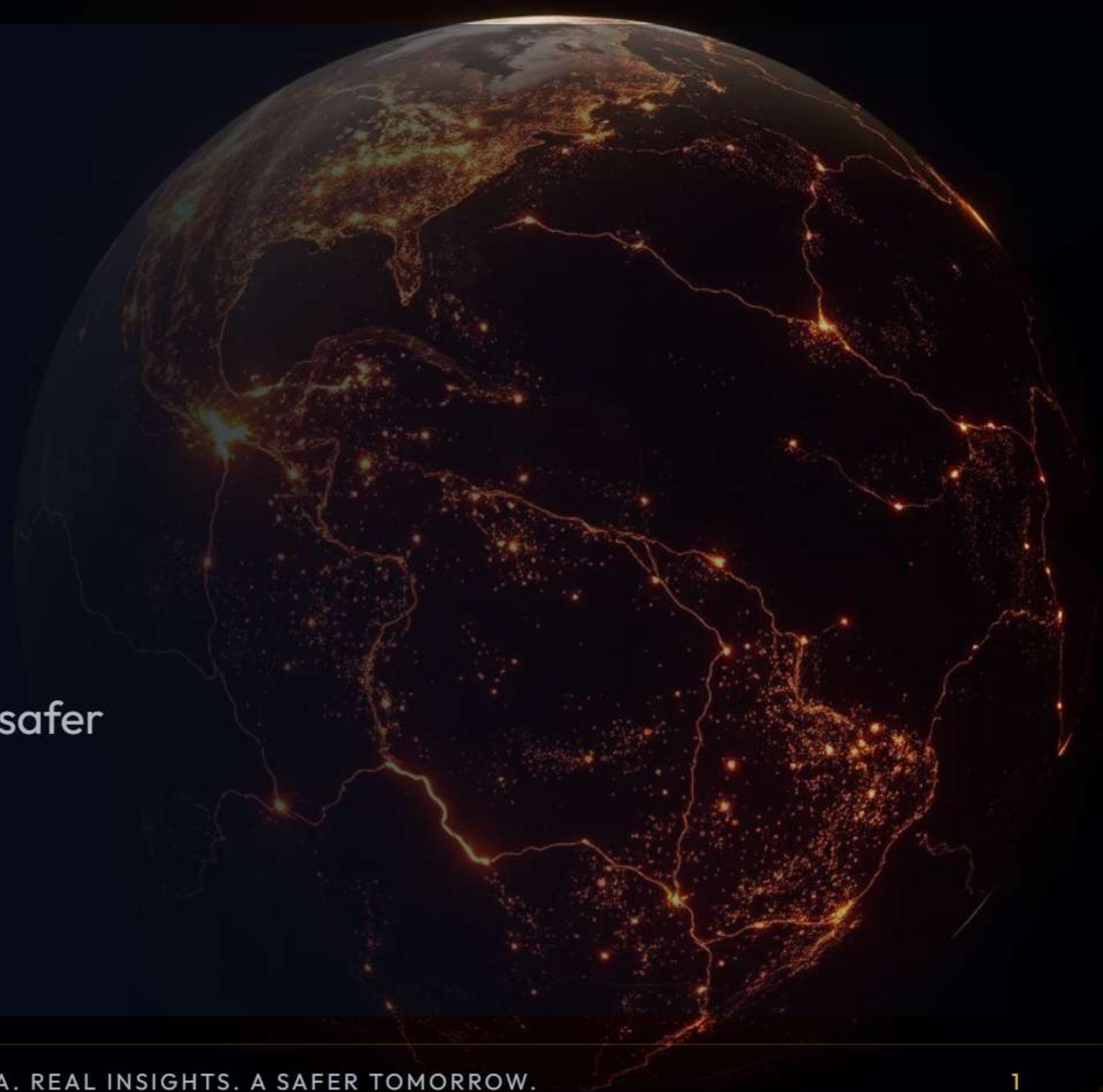
Turning earthquake data into insights for a safer tomorrow.

PRESENTED BY

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The Challenge: From Chaos to Clarity



Earthquakes generate massive, unstructured GeoJSON data.



Manual analysis is slow, error-prone, and lacks real-time insight.



Stakeholders need immediate risk assessment for disaster response.



Goal: turn raw data into actionable intelligence, automatically.

- ◆ **MASSIVE DATA**
- ◆ **COMPLEX FORMATS**
- ◆ **TIME CRITICAL**
- ◆ **LIVES AT STAKE**

Project Objectives



Automate Ingestion

Fetch 390+ days of earthquake data.



Ensure Data Quality

Clean, validate, and deduplicate.



Enable Analytics

Create business-ready Gold tables.



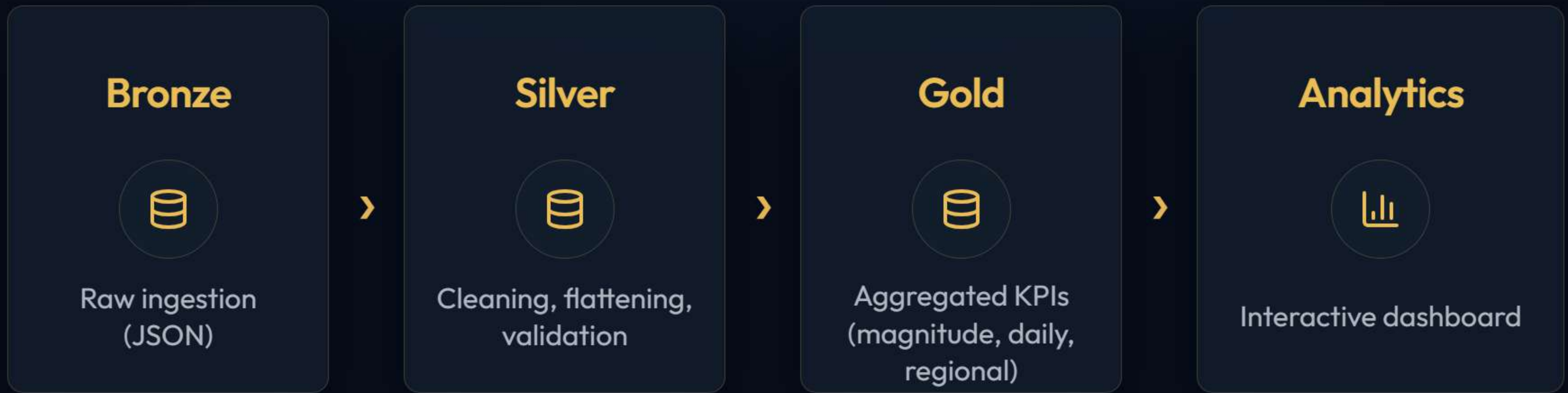
Visualize Insights

Build an interactive dashboard.



CLEAN DATA
CLEAR INSIGHTS
BIGGER IMPACT

The Engine: Medallion Architecture



Technologies Used



Data Ingestion & Processing
PySpark



Data Storage
Delta Lake



Geospatial Data Source
Earthquake API



Analytics & Visualization
Dashboard



Automation
Scheduled pipelines



Version Control
Git & GitHub

Key Insights & Features



Global Heatmap

Real-time epicenter visualization.



Trend Analysis

Magnitude frequency over time.



Depth Correlation

Shallow vs. deep patterns.



Regional Risk

Top 10 most active countries.



Live KPIs

Total events, max magnitude, average depth.



Automation & Pipeline



Scalable

Delta Lake ensures ACID compliance.



Reliable

Automated, monitored, and resilient.

Data Quality & Governance



Validation

Coordinate bounds checked (-180 to 180).



Deduplication

Unique event_id enforced.



Null Handling

Critical fields imputed or flagged.



Time Travel

Delta Lake allows historical queries.



CLEAN DATA
TRUSTED INSIGHTS
BETTER DECISIONS



Faster Decision Making

Real-time vs. manual.



Risk Mitigation

Identify high-risk zones instantly.



Scalable Solution

Extensible to other geospatial data.



Open Source

Fully documented and reproducible.

INSIGHTS THAT SAVE LIVES

Thank You



Together, we turn earthquake data into a safer tomorrow.

“More data. Safer communities. A stronger tomorrow.”

 github.com/yourusername/SOBZ-Earthquake-Analytics

 your.email@example.com

 LinkedIn Profile